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REPUBLIC POLYTECHNIC

E211 – Operations Planning II

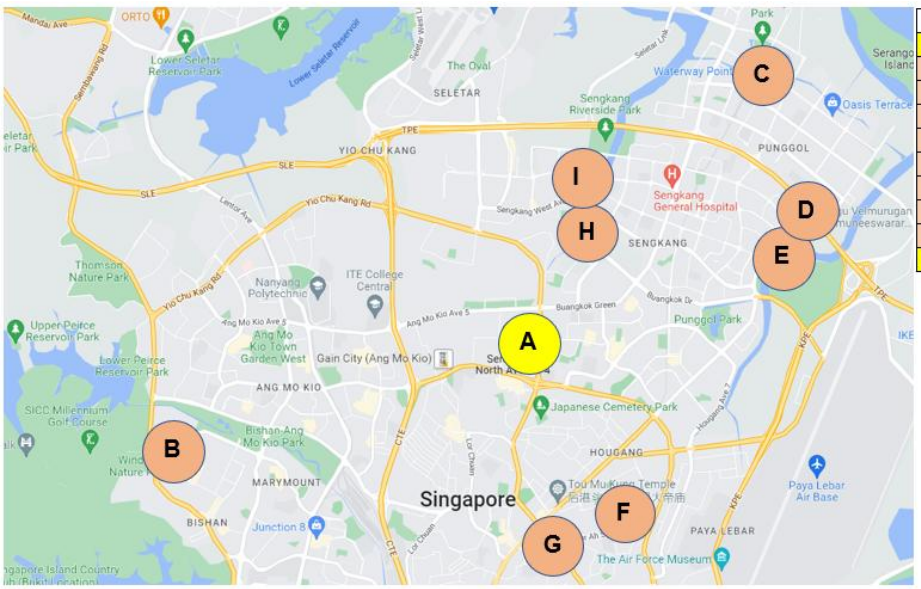
Lesson 18 Worksheet

Problem Statement

WantTheToys Pte Ltd is a Singapore based company that produces educational toys. To carry out the toy distribution in Singapore, WantTheToys is working with ExpressD, a last-mile delivery company.

Every day, ExpressD will deliver products from WantTheToys' warehouse to fulfil customer orders. Drivers often spend time planning their delivery routes as customer locations change daily. To complicate matters, the quantity of products to deliver would sometimes exceed capacity of the van. Also, some customers might request deliveries to be made at specific times. Is there a way to help ExpressD plan their delivery routes efficiently?

Use the data below for the exercise.



Label	Address
A (Depot)	Serangoon North
B	Jalan Angklong
C	Sumang link
D	Rivervale Crescent
E	Rivervale Street
F	Jalan Pelikat
G	Lorong Lew Lian
H	Fernvale Link
I	Sengkang West Way
A (Depot)	Serangoon North

Figure 1: Depot and Delivery Locations

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Table 1: Distance in Kilometers Between Locations

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	**A**	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	**B**	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	**C**	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	**D**	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	**E**	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	**F**	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	**G**	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	**H**	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	**I**	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

Table 2: Customer Demand

Customer	Delivery Load (kg)
B	100
C	100
D	200
E	300
F	80
G	70
H	180
I	70

1. Determine a route for the deliveries to the 8 customers using the Nearest Neighbour method.
   Route (tour): A – G – F – E – D – C – I – H – B – A

Route (tour):

From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1
	Total Dist.	**42.5 km**

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2. Determine a route for the deliveries using Clarke and Wright Savings method. You may use the attached template for workings.

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L18 Templates.xlsx

Rank	Saving	From	To	Tour	Remarks
1	9.5	D	E	**(1) A-D-E-A**	Initiate new tour
2	8.3	C	D	**(1) A-C-D-E-A**	Join C and D
3	7.6	C	E		C and E are already assigned to the same route
4	6.7	H	I	**(1) A-C-D-E-A** **(2) A-H-I-A**	Initiate new tour
5	5.8	D	I		D is not adjacent to A
6	5.8	F	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-A**	Initiate new tour
7	5.6	D	H		D is not adjacent to A
8	5.2	B	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-B-A**	Join B and G
9	5.0	E	H	**(1) A-C-D-E-H-I-A** **(2) A-F-G-B-A**	Join E and H; Merge the 2 tours
10	4.7	E	I		E and I are already assigned to the same route
11	4.4	C	I		C and I are already assigned to the same route
12	3.8	C	H		C and H are already assigned to the same route
13	3.5	D	F		D is not adjacent to A
14	3.3	B	F		B and F are already assigned to the same route
15	3.0	E	F		E is not adjacent to A
16	2.8	B	I	**(1) A-C-D-E-H-I-B-G-F-A**	Join B and I; Merge the 2 tours
17	2.7	D	G		D and G are already assigned to the same route
18	2.6	B	H		B and H are already assigned to the same route
19	1.9	E	G		E and G are already assigned to the same route
20	1.2	C	F		C and F are already assigned to the same route
21	1.2	G	I		G and I are already assigned to the same route
22	0.6	B	C		B and C are already assigned to the same route
23	0.6	F	H		F and H are already assigned to the same route
24	0.6	G	H		G and H are already assigned to the same route
25	0.1	C	G		C and G are already assigned to the same route
26	0.1	F	I		F and I are already assigned to the same route
27	-0.3	B	D		B and D are already assigned to the same route
28	-1.0	B	E		B and E are already assigned to the same route

Route (tour): A – C – D – E – H – I – B – G – F – A

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist.	**39.5**	**km**

3. Compare the routes derived from both methods above. Which is a better route? In general, will any of these methods always provide optimal routing solutions?
   Clarke and Wright Savings is a better route. These methods do not guarantee optimal routing solutions.

4. Suppose ExpressD would like to assign 2 vans to provide delivery to the 8 customers. Discuss the approach you could use to plan the assignment of van and the route for each van to take.
   A-C-D-E-H-I-A
   A-F-G-B-A

5. Capacity of ExpressD's vans is **600 kg**. Given the delivery load for each customer below, answer the following questions:

Customer	Delivery Load (kg)
B	150
C	100
D	150
E	200
F	80
G	100
H	180
I	90

a. Are the routes that you worked out previously still feasible? Why?
   (Nearest Neighbour Method):
   Route (tour): A – G – F – E – D – C – I – H – B – A
   100+80+200+150+100+90+180+150=1050

   (Clarke & Wright Savings Method):
   Route (tour): A – C – D – E – H – I – B – G – F – A
   100+150+200+180+90+150+100+80=1050

   The routes are not feasible as total delivery load (1050kg) > capacity (600kg)

b. How should the company plan the delivery routes now? Work out a new solution using Clark and Wright savings method.
   The company should consider the constraints of total delivery load of each customer and total van capacity. Total load should not exceed total capacity.

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Rank	Saving	From	To	Tour	Cumulative Load	Remarks
1	9.5	D	E	**(1) A-D-E-A**	**150+200=350**	Initiate a new route: A-D-E-A
2	8.3	C	D	**(1) A-C-D-E-A**	**100+150+200=450**	Link C with D
3	7.6	C	E			C and E are already assigned to the same route
4	6.7	H	I	**(1) A-C-D-E-A** **(2) A-H-I-A**	**100+150+200=450** **180+90 = 270**	Initiate another route: A-H-I-A
5	5.8	D	I			D is not adjacent to A
6	5.8	F	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-A**	**100+150+200=450** **180+90 = 270** **80+100=180**	Initiate another route: A-F-G-A
7	5.6	D	H			D is not adjacent to A
8	5.2	B	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-B-A**	**100+150+200=450** **180+90 = 270** **80+100+150=330**	Link B with G
9	5.0	E	H	**(1) A-C-D-E-I-A** **(2) A-H-A** **(3) A-F-G-B-A**	**100+150+200+90=540, 180** **80+100+150=330**	Cannot link E with H, because truck capacity (600 kg) will be violated, start new route for H.
10	4.7	E	I			Both E and I are already in a route
11	4.4	C	I			Both C and I are already in a route
12	3.8	C	H			Cannot merge C with H, because truck capacity (600 kg) will be violated
13	3.5	D	F			D is not adjacent to A
14	3.3	B	F			Both B and F are already in a route
15	3.0	E	F			E is not adjacent to A
16	2.8	B	I	**CWS Original route:** **A-C-D-E-H-I-B-G-F-A** **New route (With Vehicle Capacity):** **(1) A-C-D-E-A** **(2) A-H-I-B-G-F-A**	**100+150+200=450,** **180+90+150+100+80=600**	Link B and I
17	2.7	D	G			
18	2.6	B	H			
19	1.9	E	G			
20	1.2	C	F			
21	1.2	G	I			
22	0.6	B	C			
23	0.6	F	H			
24	0.6	G	H			
25	0.1	C	G			
26	0.1	F	I			
27	-0.3	B	D			
28	-1.0	B	E			
Truck 1: A-C-D-E-A with load 450kg and total distance 16.6km **Truck 2: A-H-I-B-G-F-A with load 600kg and total distance 27.9km**						

Truck 1

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	A	5.2
	Total Dist.	**16.6** **km**

Truck 2

From	To	Distance
A	H	4
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3
	Total Dist.	**27.9** **km**

Routes (tour):

Truck 1: A-C-D-E-A

Truck 2: A-H-I-B-G-F-A

6. If some customers request for the delivery to be done at certain time periods, how should ExpressD consider this in the planning of the route?
   ExpressD should consider the constraints of customers:
   - Time windows (timing open for receiving of goods)
   - Priority
   - Pickup and delivery

7. Discuss the challenges and issues that may cause delivery failure or unexpected delays in deliveries.
   Delivery failures often occur because the address is incorrect, incomplete, or ambiguous. This requires redelivery attempts or returns to the depot, further delaying the process. Shortages of drivers, warehouse staff, or other operational personnel, especially during peak seasons, can slow down sorting, loading, and deliveries.